

Trajectory Planning in UAV Communication with Jamming

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Abstract—Unmanned aerial vehicles (UAVs) have attracted a lot of attention in wireless communications due to their high mobility. However, the jamming resistance by exploiting the UAV's mobility is still an open problem in UAV communication. In this paper, we investigate the deployment and trajectory planning problem in three-dimensional (3D) space with jamming, where a UAV communicates with the ground node while some jammers attack the transmission. The formulated optimization is non-convex and hard to be solved directly. By introducing the slack variables and leveraging the successive convex approximation, we design a trajectory planning method to optimize the UAV's horizontal position and vertical altitude in 3D space. Furthermore, we study the static deployment scenario, which also provides an upper bound for the trajectory planning case with unlimited speed. For single jammer case, we derive an analytical expression for the optimal UAV's location to efficiently resist attacks. For multiple jammers case, we provide a sub-optimal deployment scheme. Simulation results show that the proposed trajectory planning method and deployment scheme outperform the benchmark scheme.

Index Terms—Deployment, jamming, trajectory planning, unmanned aerial vehicles.

I. INTRODUCTION

The increasing applications of UAVs in emergency communications and enhanced coverage have pushed the UAV communications to be an important and interesting research topic [1], [2]. Besides the efforts on air-to-ground channel model [3], many recent studies have focused on the use cases of UAVs as aerial base stations [4], [5], mobile relays [6], [7], energy sources [8], [9], etc. [10]. It has been shown firstly in [1], [6] that employing the UAV to assist the wireless communications can bring significant performance gains by exploiting the channel variations. Furthermore, the authors in [10] investigate the power control problem in UAV-supported ultra dense networks to achieve this gain. Other UAV-related researches include UAV-aid Internet of Things [11], cache [12]. Nevertheless, these mentioned studies have not considered the communication case with jamming, which will greatly degrade the system performance.

Due to the broadcast nature of wireless transmission, UAV systems easily suffer from jamming. Recently, there are some emerging researches on the UAV-involved communications with jamming [13], [14]. In [13], the prospect theory are applied to enhance the UAV transmission with smart attacks.

The authors in [14] aim to improve the anti-jamming communication in vehicular ad-hoc networks (VANETs) by utilizing UAVs to relay the message of onboard units. A UAV relay strategy is proposed without requiring the prior information of the jamming and UAV channel model. These efforts in [13], [14] build the foundations for researches on UAV-involved communications with jamming. However, to our best knowledge, the jamming resistance by exploiting the UAV's mobility is still an open problem in UAV communication.

Motivated by the above observations, we investigate the deployment and trajectory planning problem in UAV communication with jamming, where a UAV communicates with the ground node while some jammers attack this transmission. By introducing the slack variables and leveraging the successive convex approximation (SCA), we first design a trajectory planning method to optimize the UAV's horizontal position and vertical altitude in three-dimensional (3D) space. Then, we analyze the achievable throughput for static deployment with jamming. The main contributions are as follows:

- We propose a trajectory planning method in 3D space. Different from most existing studies that optimize the UAV's trajectory mainly from the horizontal position perspective, the UAV's altitude is also considered in our investigated problem to enhance the anti-jamming performance.
- We design an anti-jamming approach by dynamically adjusting the UAV's trajectory. Notice that the proposed approach can be extended to other scenarios, such UAV communications with obstacles.
- We derive an analytical expression for the optimal UAV's location to efficiently resist single jammer. For multiple jammers, we provide a suboptimal deployment scheme, which provides an upper bound for trajectory planning case.

The rest of this paper is organized as follows. In Section II, we illustrate the system model and formulate the optimization problem. Then, we design a trajectory planning method in Section III. In Section IV, the achievable throughput is analyzed for static deployment case with jamming. Simulation results are presented to verify the effectiveness of the proposed approach in Section V. Finally, we conclude the paper in

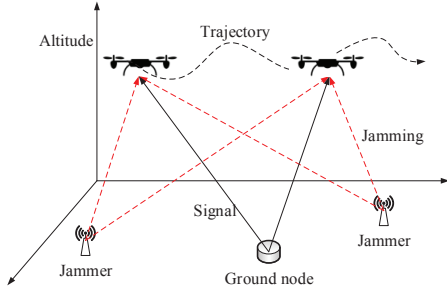


Fig. 1. The investigated scenario where the UAV communicates with a ground node with jamming.

Section VI.

II. SYSTEM MODEL AND PROBLEM FORMULATION

Consider a scenario where the UAV communicates with a ground node within time length T . As an example, in vehicular networks, a mobile UAV aims to receive transport and surveillance information from the ground unit [15]. Meanwhile, a set of ground jammers, denoted by $\mathcal{M} = \{1, 2, \dots, m, \dots, M\}$, proactively attacks the UAV communication, as shown in Fig. 1. The time length of interest T is divided into $\mathcal{N} = \{1, 2, \dots, n, \dots, N\}$ time slot with each length δ_t , i.e., $T = N\delta_t$. Thus, the UAV's trajectory is given as $\{w[n]\}$, and $w[n] = \{x[n], y[n], z[n]\}$ is the UAV's location at n -th time slot in 3D space. The $z[n]$ represents the UAV's altitude, which may vary over the time slot to resist the jamming. The locations of the ground node and m -th jammer are fixed, given as $w_c = \{x_c, y_c, 0\}$ and $w_m = \{x_m, y_m, 0\}$, respectively.

In this paper, we consider a proactive anti-jamming planning method, i.e., in advance designing the UAV's trajectory within the future T time length. In a long planning period of time, future instantaneous channel state information (CSI) and transmit powers $\{p_m\}$ of jammers are unpredictable accurately although there are some estimation techniques. Therefore, the planning are based on statistical information. The channel power gain between the UAV and ground node (m -th jammer) is dominated by LoS (line-of-sight) transmission and given by:

$$g_m[n] = \beta_0 \|w[n] - w_m\|^{-\alpha}, \quad (1)$$

at n -th time slot, where β_0 is the channel power gain at the reference distance d_0 , α is the path loss exponent, $\|a\|$ provides the Euclidean norm for vector a , i.e., $\|w[n] - w_m\| = \sqrt{(x[n] - x_m)^2 + (y[n] - y_m)^2 + (z[n] - z_m)^2}$.

The achievable throughput at n -th time slot is given by

$$R[n] = \log_2 \left(1 + \frac{p_0 g_0[n]}{\sum_{m=1}^M p_m g_m[n] + \sigma^2} \right), \quad (2)$$

where p_0 is the transmit power of ground node, p_m is the statistical value of m -th jammer's transmit power, and σ^2 is the noise power. Our aim is to maximize the sum throughput received by the UAV with jamming signals, via designing the

UAV's trajectory in 3D space. Mathematically, the problem can be formulated as follows:

$$\begin{aligned} \max_{\{w[n]\}} R &= \sum_{n=1}^N R[n] \\ \text{s.t. } C1 : &\|w[n] - w[n-1]\| \leq V\delta_t, \forall n, \\ C2 : &H_{\min} \leq z[n] \leq H_{\max}, \forall n, \end{aligned} \quad (3)$$

where V is the maximum speed, $H_{\min}$ and $H_{\max}$ correspond to the minimum and maximum allowable flying altitude, $w[0] = \{x[0], y[0], z[0]\}$ and $w[N] = \{x[N], y[N], z[N]\}$ are the start and end locations of the UAV. $C1$ and $C2$ are the trajectory and altitude constraints, respectively. In practice, some smart jammers can adapt their transmit powers in real time by observing the UAV and ground node's behaviors. In this case, the UAV and ground node's transmit powers can be further optimized, which is a classical water filling problem with a given trajectory.

III. TRAJECTORY PLANNING METHOD WITH JAMMING

The formulated problem in (3) requires optimizing the UAV's trajectory, which is difficult to be directly addressed due to: i) there are interference signals from jammers, which vary with the movement of the UAV; ii) the objective is non-convex. To this end, we address this problem by introducing the slack variables and leveraging the SCA.

Theorem 1: Introducing slack variables $\{S[n]\}$ and $\{I[n]\}$, an optimization problem equivalent to the problem in (3) can be obtained as follows:

$$\begin{aligned} \max_{\{w[n], S[n], I[n]\}} &\sum_{n=1}^N \log_2 \left(1 + \frac{S[n]^{-1}}{I[n]} \right) \\ \text{s.t. } &C1, C2, \\ C3.1 : &\begin{cases} p_0 \beta_0 \|w[n] - w_0\|^{-\alpha} \geq S[n]^{-1}, \forall n, \\ S[n] \geq 0, \forall n, \end{cases} \\ C4.1 : &\sum_{m=1}^M p_m \beta_0 \|w[n] - w_m\|^{-\alpha} + \sigma^2 \leq I[n], \forall n. \end{aligned} \quad (4)$$

Proof: It can be proved by contradiction. Specifically, if there are $p_0 \beta_0 \|w[n] - w_0\|^{-\alpha} = S[n]^{-1}, \forall n$ and $\sum_{m=1}^M p_m \beta_0 \|w[n] - w_m\|^{-\alpha} + \sigma^2 = I[n], \forall n$, the solutions to the problems in (3) and (4) are the same. Assume that the conditions are not strictly true, i.e., there exists $n \in \mathcal{N}$ such that $S[n]^{-1} < p_0 \beta_0 \|w[n] - w_0\|^{-\alpha}$ or $I[n] > \sum_{m=1}^M p_m \beta_0 \|w[n] - w_m\|^{-\alpha} + \sigma^2$. Then, the objective can be always improved by increasing $S[n]^{-1}$ or decreasing $I[n]$ such that the inequalities are strictly positive. Thus, the optimal solution to the problem in (4) satisfies the equalities. ■

The constraint in $C3.1$ can be further rewritten as the following convex constraint:

$$C3.2 : \begin{cases} \|w[n] - w_0\|^\alpha - p_0 \beta_0 S[n] \leq 0, \forall n, \\ S[n] \geq 0, \forall n. \end{cases} \quad (5)$$

Lemma 1: The objective function in (4) is lower bounded at point $(S^k[n], I^k[n])$ by:

$$R_l(S[n], I[n]) = \log_2(1 + 1/(S^k[n] I^k[n])) + A(S[n] - S^k[n]) + B(I[n] - I^k[n]), \quad (6)$$

where $A = -\frac{\log_2 e}{S^k[n] + (S^k[n])^2 I^k[n]}$ and $B = -\frac{\log_2 e}{I^k[n] + (I^k[n])^2 S^k[n]}$.

Proof: It can be easily proved that $f(x, y) = \log_2(1 + 1/xy)$ is a convex function. Since the first-order Taylor approximation of a convex function is a global under-estimator, there is

$$\log_2\left(1 + \frac{1}{xy}\right) \geq R_l = \log_2\left(1 + \frac{1}{x^k y^k}\right) - \frac{\log_2 e}{x^k + (x^k)^2 y^k} (x - x^k) - \frac{\log_2 e}{y^k + (y^k)^2 x^k} (y - y^k), \quad (7)$$

at point (x^k, y^k) . Let $x = S[n], y = I[n]$, the Lemma 1 is proved. ■

Thus, the problem in (4) is lower bounded by the following optimization at given point $(S^k[n], I^k[n])$:

$$\begin{aligned} \max_{\{w[n], S[n], I[n]\}} & R_l(S[n], I[n]) \\ \text{s.t. } & C1, C2, C3.2, C4.1. \end{aligned} \quad (8)$$

Moreover, it can be observed that $R_l(S[n], I[n])$ can be simplified as follows:

$$\begin{aligned} R_l(S[n], I[n]) &= A * S[n] + B * I[n] + Con \\ &\triangleq R_l^s(S[n], I[n]) + Con, \end{aligned} \quad (9)$$

where

$$\begin{aligned} Con &= \log_2\left(1 + \frac{1}{S^k[n] I^k[n]}\right) - A * S^k[n] - B * I^k[n], \\ R_l^s(S[n], I[n]) &= A * S[n] + B * I[n]. \end{aligned} \quad (10)$$

Since the second term (Con) in (9) is independent of the investigated variables $\{w[n], S[n], I[n]\}$, the objective in (8) can be substitute with $R_l^s(S[n], I[n])$. However, this still is not a convex optimization. To this end, we introduce the Theorem 2.

Theorem 2: By introducing variables $\{d_m[n]\}$, the problem in (8) can be resolved via addressing the following optimization:

$$\begin{aligned} \max_{\substack{\{w[n], S[n]\} \\ \{I[n], d_m[n]\}}} & R_l^s(S[n], I[n]) \\ \text{s.t. } & C1, C2, C3.2, \\ & C4.2 : \begin{cases} \sum_{m=1}^M p_m \beta_0 d_m[n]^{-\alpha/2} + \sigma^2 \leq I[n], \forall n, \\ d_m[n] \leq \|w[n] - w_m\|^2, \forall n, m, \\ d_m[n] \geq 0, \forall n, m. \end{cases} \end{aligned} \quad (11)$$

Proof: It can be proved by following the proof of Theorem 1. Due to the paper limit, we omit it in this paper. ■

Algorithm 1 SCA-based trajectory planning method

- 1: Initialize $\{w[n], S[n], I[n], d_m[n]\}^0$ and $k = 1$
 - 2: **Repeat**
 - 3: Solve the problem in (14) via standard convex optimization techniques and obtain the optimal solution $\{w[n], S[n], I[n], d_m[n]\}^*$
 - 4: Update the variable $\{w[n], S[n], I[n], d_m[n]\}^k = \{w[n], S[n], I[n], d_m[n]\}^*$
 - 5: Update $k = k + 1$
 - 6: **Until** some termination conditions are met
-

The second constraint in C4.2 is non-convex. However, it can be efficiently handled by deriving the global lower bound. Denote $F_m[n]|_{w[n]} = \|w[n] - w_m\|^2$. Then, we have

$$\begin{aligned} F_m[n]|_{w[n]} &= (x[n] - x_m)^2 + (y[n] - y_m)^2 + z^2[n] \\ &= x^2[n] + y^2[n] + z^2[n] \\ &\quad - 2x_m x[n] - 2y_m y[n] + x_m^2 + y_m^2. \end{aligned} \quad (12)$$

Since $x^2[n], y^2[n], z^2[n]$ are convex,

$$\begin{aligned} F_m[n]|_{w[n]} &\geq L_m[n] \\ &= 2x^k[n] x[n] - (x^k[n])^2 + 2y^k[n] y[n] - (y^k[n])^2 + x_m^2 \\ &\quad + 2z^k[n] z[n] - (z^k[n])^2 - 2x_m x[n] - 2y_m y[n] + y_m^2. \end{aligned} \quad (13)$$

Therefore, the problem in (4) is lower bounded by the following convex optimization:

$$\begin{aligned} \max_{\substack{\{w[n], S[n]\} \\ \{I[n], d_m[n]\}}} & R_l^s(S[n], I[n]) \\ \text{s.t. } & C1, C2, C3.2, \\ & C4.3 : \begin{cases} \sum_{m=1}^M p_m \beta_0 d_m[n]^{-\alpha/2} + \sigma^2 \leq I[n], \forall n, \\ d_m[n] \leq L_m[n], \forall n, m, \\ d_m[n] \geq 0, \forall n, m, \end{cases} \end{aligned} \quad (14)$$

which can be solved by standard convex optimization techniques, such as interior point method [16]. Finally, the overall procedure for solving the optimization problem in (3) is presented in **Algorithm 1**.

Convergence and Complexity Analysis: Since the k -th solution to (14) is feasible to $(k+1)$ -th optimization, the resulting values of (14) are non-decreasing. Moreover, it must be upper bounded by the optimal solution to (3), the convergence is guaranteed. The main complexity of the proposed trajectory planning method lies in step 3, which require solving a series of convex problems. Thus, the computational complexity is about $O(K(5N + MN)^3)$, where K is the iteration number, M is the number of jammers.

IV. STATIC DEPLOYMENT ANALYSIS

In this section, we analyze the static deployment of the UAV with jamming. Specifically, the aim is to find the optimal

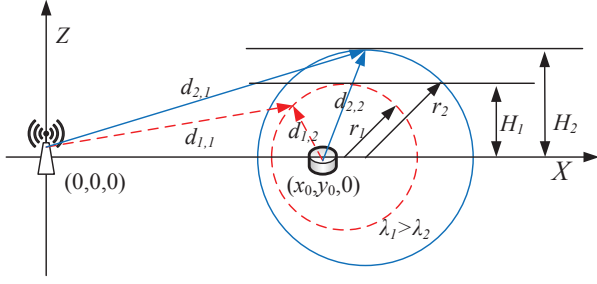


Fig. 2. Illustration of the UAV's location with single jammer.

UAV's location w that maximizes the sum throughput as follow:

$$\max_w \log_2 \left(1 + \frac{p_0 \beta_0 \|w - w_0\|^{-\alpha}}{\sum_{m=1}^M p_m \beta_0 \|w - w_m\|^{-\alpha} + \sigma^2} \right) \quad (15)$$

s.t. C2.

It can be observed that the problem in (15) is an asymptotic case of problem in (3) with unlimited speed $V \rightarrow \infty$. Therefore, the solution to (15) provides an upper bound for (3). The optimal solution to (15) is hard to be achieved due to its non-convex property. Next, we propose a sub-optimal deployment method, which can be employed with low computational complexity.

Since the transmit powers of jammers are usually much stronger than that of noise, the problem can be approximated by

$$\max_w \frac{\|w - w_0\|^{-\alpha}}{\sum_{m=1}^M p_m \|w - w_m\|^{-\alpha}}. \quad (16)$$

Using the reciprocal of (16), there is

$$\max_w \frac{\|w - w_0\|^{-\alpha}}{\sum_{m=1}^M p_m \|w - w_0\|^{-\alpha}} = \min_w \sum_{m=1}^M p_m \frac{\|w - w_m\|^{-\alpha}}{\|w - w_0\|^{-\alpha}}. \quad (17)$$

Next, we first study the single jammer case $M = 1$, which provides sight for multiple jammers case.

Theorem 3: For any jammer $w_m = \{x_m, y_m, 0\}$ and ground node $w_0 = \{x_0, y_0, 0\}$, the UAV's optimal location is $w_m^o = \left(\frac{\lambda_1^2 x_0 - x_m}{\lambda_1^2 - 1}, \frac{\lambda_1^2 y_0 - y_m}{\lambda_1^2 - 1}, H_{\min} \right)$ with $\lambda_1 = \frac{\|w_0 - w_m\|}{2H_{\min}} + \sqrt{1 + \left(\frac{\|w_0 - w_m\|}{2H_{\min}} \right)^2}$.

Proof: Considering m -th jammer and ground node, the optimization in (17) is

$$\max_w \frac{\|w - w_m\|}{\|w - w_0\|}, \quad (18)$$

where the objective is the ratio of distances from the UAV to the jammer and ground node, respectively. Denote the ratio

$$\lambda = \frac{\|w - w_m\|}{\|w - w_0\|} = \frac{\sqrt{(x - x_m)^2 + (y - y_m)^2 + z^2}}{\sqrt{(x - x_0)^2 + (y - y_0)^2 + z^2}}. \quad (19)$$

By arranging the above expression, we have

$$x^2 + 2 \frac{x_m - \lambda^2 x_0}{\lambda^2 - 1} x + \frac{\lambda^2 x_0^2 - x_m^2}{\lambda^2 - 1} + y^2 + 2 \frac{y_m - \lambda^2 y_0}{\lambda^2 - 1} y + \frac{\lambda^2 y_0^2 - y_m^2}{\lambda^2 - 1} + z^2 = 0. \quad (20)$$

Further, it can be written as

$$\left(x + \frac{x_m - \lambda^2 x_0}{\lambda^2 - 1} \right)^2 + \left(y + \frac{y_m - \lambda^2 y_0}{\lambda^2 - 1} \right)^2 + z^2 = \frac{(x_0 - x_m)^2 + (y_0 - y_m)^2}{(\lambda - 1/\lambda)^2}. \quad (21)$$

This means all the points that make the ratio as λ form a sphere, with $\left(\frac{\lambda^2 x_0 - x_m}{\lambda^2 - 1}, \frac{\lambda^2 y_0 - y_m}{\lambda^2 - 1}, 0 \right)$ as the center of the sphere. Since maximizing the objective results in $\lambda > 1$, the radius of the sphere is $r = \frac{\sqrt{(x_0 - x_m)^2 + (y_0 - y_m)^2}}{(\lambda - 1/\lambda)}$. It can be observed that larger objective λ means smaller radius r , and vice versa. The smallest radius is given as $r = H_{\min}$ since the UAV's altitude is lower bounded by $H_{\min}$. Fig. 2 presents an illustration of the UAV's location with single jammer, where the locations of jammer and ground node are respectively $(0, 0, 0)$ and $(x_0, y_0, 0)$, $\lambda_1 = d_{1,1}/d_{1,2} > \lambda_2 = d_{2,1}/d_{2,2}$. It can be seen from Fig. 2 that $\max \lambda = \min r = H_{\min}$.

Let $r = H_{\min}$, we can get $\lambda_1 = \frac{\|w_0 - w_m\|}{2H_{\min}} + \sqrt{1 + \left(\frac{\|w_0 - w_m\|}{2H_{\min}} \right)^2} > 0$ and $\lambda_2 = \frac{\|w_0 - w_m\|}{2H_{\min}} - \sqrt{1 + \left(\frac{\|w_0 - w_m\|}{2H_{\min}} \right)^2} < 0$. Since there is $\lambda > 0$, $\lambda = \lambda_1$. Therefore, the UAV's optimal location is $w_m^o = \left(\frac{\lambda_1^2 x_0 - x_m}{\lambda_1^2 - 1}, \frac{\lambda_1^2 y_0 - y_m}{\lambda_1^2 - 1}, H_{\min} \right)$. ■

For multiple jammers case, the arithmetic mean $w^o = \frac{1}{M} \sum_{m=1}^M w_m^o$ can be used as a sub-optimal solution to (15) since its optimal solution cannot be efficiently obtained. As a result, the achievable throughput is given as

$$R^s = \log_2 \left(1 + \frac{p_0 \beta_0 \|w^o - w_0\|^{-\alpha}}{\sum_{m=1}^M p_m \beta_0 \|w^o - w_m\|^{-\alpha} + \sigma^2} \right). \quad (22)$$

V. SIMULATIONS AND DISCUSSIONS

In this section, we show the UAV's behavior with jamming and demonstrate the effectiveness of the proposed algorithm by conducting in-depth simulations. Consider a $500 \times 500 \times 150$ m³ scenario with a ground node (located at $(400, 300, 0)$) and two jammers (located at $(200, 150, 0)$ and $(400, 280, 0)$, respectively). The system parameters are set as follows: $H_{\min} = 50$ m, $H_{\max} = 150$ m, $V = 60$ m/s, $p_m = 0.4$ W, and $\delta_t = 0.5$ s. The noise power spectrum density is -169 dBm/Hz and the considered bandwidth is 1 MHz. The channel power gain at the reference distance $d_0 = 1$ m is $\beta_0 = -30$ dB. Unless otherwise specified, the transmit power of the ground node is $p_0 = 0.1$ W.

Fig. 3 illustrates the UAV's trajectory in 3D space, where “*”, “△”, “★”, and “×” represent the UAV's start (end) location, jammer, ground node, and deployment point, respectively. The dotted line is the projection of the trajectory

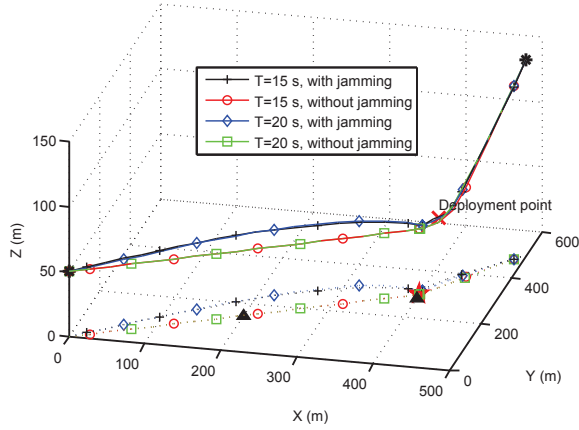


Fig. 3. The UAV's trajectory in 3D space with and without jammer, where the term “*”, “△”, “*”, and “x” represent the UAV's start (end) location, jammer, ground node, and deployment point, respectively. The dotted line is the projection of the trajectory on the horizontal plane.

on the horizontal plane. The considered time lengths are 15 s and 20 s. For comparison, we also present the trajectory without jamming. It can be first observed that when there is no jamming, the UAV directly flies towards the ground node from the start location. In the presence of jamming, the UAV intentionally avoids the jammers, which can be seen from the projection on the horizontal plane.

To further demonstrate the UAV's behavior, Fig. 4 plots the UAV's altitude and speed over the time slot. It is found from Fig. 4(a) that while the UAV approaches the jammer, it increases its altitude to avoid the jamming, in addition to the horizontal distance away from the jammer (shown in Fig. 3). Through it all, two states for the UAV occur, i.e., flying at the maximum speed or staying over a certain point. However, the UAV hovers over different points, i.e., (400, 315, 50) and (400, 300, 50) with and without jamming, respectively. Since the static deployment point in this case is (414, 355, 50), we see that the resulting point of the proposed trajectory planning method is closer to the static deployment point.

Fig. 5 compares the sum throughput versus the transmit power of the ground node. The benchmark scheme refers to the obtained trajectory without jamming, while the static deployment case (also the scenario with unlimited speed) is regarded as an upper bound. It is observed that the proposed method outperforms the benchmark scheme, and this advantage is even more pronounced with the increasing transmit power p_0 and/or the time length T . This is expected since the UAV's hovering points are different with and without jamming. The UAV can receive much more data hovering over (400, 315, 50).

VI. CONCLUSION

In this paper, we investigated the deployment and trajectory planning problem in UAV communication with jamming, where a UAV communicates with the ground node while some jammers attack this transmission. By introducing the slack variables and leveraging the successive convex approximation

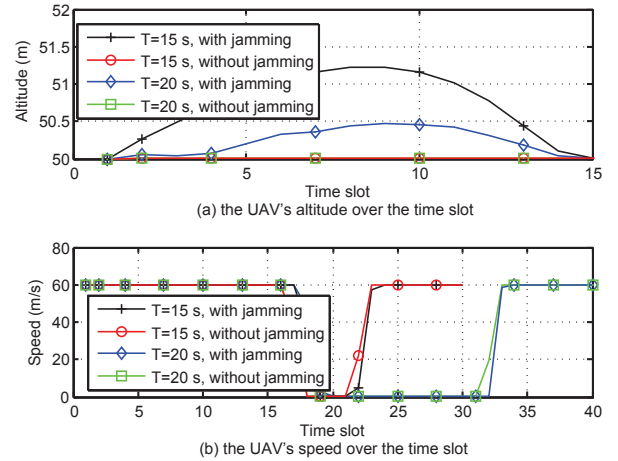


Fig. 4. The UAV's altitude and speed over the time slot.

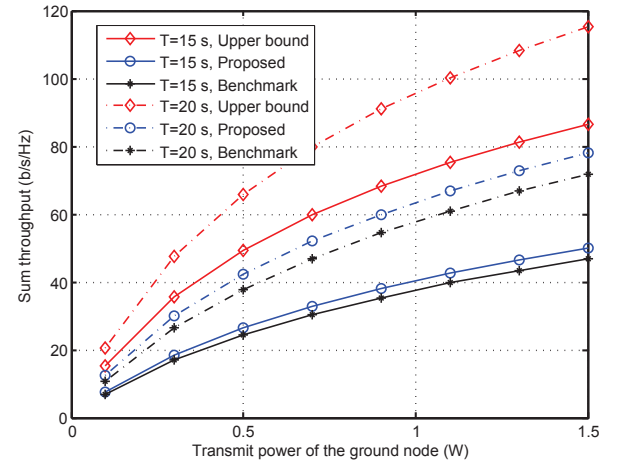


Fig. 5. The sum throughput performance versus the transmit power of the ground node.

(SCA), we designed a trajectory planning method to optimize the UAV's horizontal position and vertical altitude in 3D space. Furthermore, we analyzed the achievable throughput for static deployment. We hope that this paper shed new light on UAV communications with jamming.

VII. ACKNOWLEDGEMENT

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